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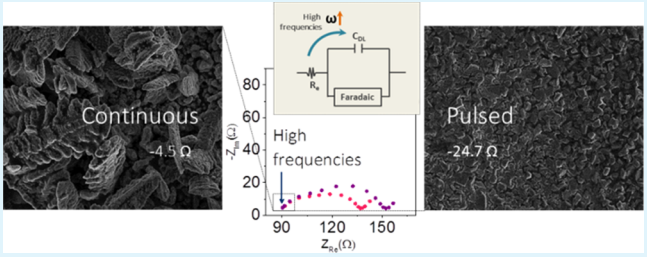
Complete Prevention of Dendrite Formation in Zn Metal Anodes by Means of Pulsed Charging Protocols

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Supporting Information

ABSTRACT: Zn metal as anode in rechargeable batteries, such as Zn/air or Zn/Ni, suffers from poor cyclability. The formation of Zn dendrites upon cycling is the key limiting step. We report a systematic study of the influence of pulsed electroplating protocols on the formation of Zn dendrites and in turn on strategies to completely prevent Zn dendrite formation. Because of the large number of variables in electroplating protocols, a scanning droplet cell technique was adapted as a high-throughput methodology in which a descriptor of the surface roughness can be in situ derived by means of electrochemical impedance spectroscopy. Upon optimizing the electroplating protocol by controlling nucleation, zincate ion depletion, and zincate ion diffusion, scanning electron microscopy and atomic force microscopy confirmed the growth of uniform and homogenous Zn deposits with a complete prevention of dendrite growth. The implementation of pulsed electroplating as the charging protocol for commercially available Ni–Zn batteries leads to substantially prolonged cyclability demonstrating the benefits of pulsed charging in Zn metal-based batteries.



KEYWORDS: *Zinc, dendrites, batteries, electrodeposition, high-throughput approach*

1. INTRODUCTION

Although the energy density of state-of-the-art Li-ion batteries (~250 Wh kg^{-1}) is sufficient to fulfill the requirements for hybrid electric vehicles, it is still below the needs for full electric vehicles.[1] Consequently, great efforts have recently been focused on the development of a new generation of battery systems with significantly higher energy densities than those of Li-ion batteries. The use of metal anodes such as metallic lithium, magnesium, aluminum, or zinc is seen as a key step in achieving high energy densities.[2−4] Among these metal anodes, Zn possesses the highest anodic redox potential enabling its use in aqueous media whereas the others require organic solvents thus imposing severe safety issues. This fact, together with its abundance on earth, makes Zn metal anode-based batteries, for example, Zn/Ni or Zn/air, potentially low-cost systems expanding their possible application for stationary energy storage as well.[5,6] However, the intrinsic issues of Zn metal anodes, namely, hydrogen evolution, dendrite growth, and electrode passivation have prevented their implementation in commercial rechargeable batteries until now.[4,7−10]

Dendritic growth upon cycling is the main limitation as it not only leads to perforation of the separator and a short-circuit between the positive and negative electrodes, but it also enhances hydrogen evolution and electrode passivation. A number of strategies have been proposed to address this issue, which can be categorized into four groups as follows. (I) The use of special separators between electrodes acting as a physical barrier to prevent short-circuiting.[11] (II) Direct modification of the active material. For example, Parker et al. demonstrated that the use of three-dimensional structures of zinc could trap dissolved Zn species such as zincate ions ($Zn(OH)_4^{2-}$) in the vicinity of the electrode upon discharge, thus minimizing redistribution of Zn ions and shape change.[12] (III) Electrolyte additives that can be divided into metallic and organic additives. The former include metallic oxides/hydroxides such as PbO or Bi_2O_3, which are reduced to the metal prior to the reduction of $Zn(OH)_4^{2-}$ ions.[13,14] The metallic film increases the electric conductivity throughout the electrode leading to a more homogeneous current distribution, thus decreasing the shape change. Organic additives such as tetrabutylammonium bromide,[15] tetra-alkyl ammonium hydroxides,[16] triethanolamine,[17] Triton X-100,[18] or poly(ethylene glycol)[19] inhibit dendrite formation by slowing down diffusional mass transport and limiting the enhanced growth at specific sites. (IV) The electrochemical approach. In the late 1980s and early 1990s, there were several studies in which the shape change in Zn metal anodes was proposed to be mitigated by applying direct current (DC) pulses or alternating current (AC) charging protocols.[20−22] The reduction of ZnO to Zn metal proceeds via dissolution of $Zn(OH)_4^{2-}$. During the reduction, the concentration of $Zn(OH)_4^{2-}$ is depleted near the electrode surface and the reaction becomes diffusion controlled. Under

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these conditions, small heterogeneities on the electrode surface trigger the formation of dendrites as sharp features grow faster due to the enhanced mass transport due to the nonplanar diffusion (Figure 1) as compared to planar diffusion and the related mass transport to flat areas.

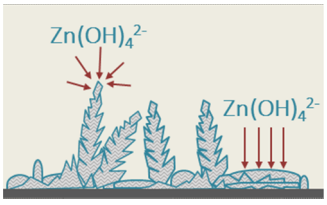


Figure 1. Schematic representation of the diffusion-limited process during Zn electroplating. Planar diffusion at the flat surface and planar plus radial diffusion at sharp features.

The application of pulsed DC or AC charging protocols aims to introduce a relaxation time after a short $Zn(OH)_4^{2-}$ reduction period to re-establish the concentration of $Zn(OH)_4^{2-}$ at the electrode surface.[23] By doing this, neighboring sites do not compete for the limited amount of $Zn(OH)_4^{2-}$ under diffusional mass transport enabling a homogeneous growth at the entire electrode surface. Although interesting results were obtained in the mentioned early studies, the complexity and the large number of interconnected parameters affecting electroplating, for example, amplitude and duration of the plating pulse, duration of the resting pulse, influence of prenucleation pulses, and so forth, discouraged sustained efforts to achieve a deeper understanding about this electrochemical approach to mitigate dendrite formation.

Although modeling and simulation have been shown to be powerful tools in predicting and accelerating the search for improved charging protocols,[31,32] enormous experimental efforts are still required due to the vast number of variables that have hindered progress, especially in the pulse charging protocol approach. Herein, we present a high-throughput procedure for the study of the pulsed charging protocol for Zn metal anode formation using a scanning droplet cell (SDC) system. This methodology allows us to analyze more than 1000 experimental conditions under very reproducible variation of the predominantly influencing parameters of the pulse protocols showing the potential of this approach to bridge the gap between modeled and experimental behaviors. Electrochemical impedance spectroscopy (EIS) was employed to derive a descriptor of the surface roughness in situ. Optimization of the charging protocol resulted in finding a set of parameters that led to the complete prevention of dendrite formation as confirmed by scanning electron microscopy (SEM) and atomic force microscopy (AFM) measurements.

2. EXPERIMENTAL SECTION

2.1. High-Throughput Study of Zn Electroplating. Metallic zinc was deposited from a solution of 12 M KOH (J. T. Baker, Netherlands) and 1 M $Zn(OH)_4^{2-}$ (ZnO, AnalaR NORMAPUR, U.K.), which was prepared using tridistilled and subsequently deionized water (Millipore, Milli-Q). Zn electroplating was carried out with a SDC coupled with a MicroAutolab/FRA potentiostat/galvanostat (Metrohm-Autolab). Note that a higher concentration of KOH, 12 M instead of 8 M, was employed to increase and decrease the concentrations of zincate and protons, respectively, minimizing the evolution of hydrogen at high current densities. The FRA modulus of the potentiostat was used for in situ EIS measurements at open circuit potential (OCP) in the frequency range of 50 kHz–1 Hz. Ag/AgCl 3 M KCl was chosen as the reference electrode and a platinum wire of 0.25 μm in diameter was used as the counter electrode. The Teflon tip (head) of the SDC had an opening of 1000 μm in diameter (area of 7.85×10^{-3} cm^2). Glassy carbon and graphite paper (SIGRAFLEX Folie F02012TH 0.20 mm × 500 mm × 50 m) were used as the working electrodes. In the case of glassy carbon, it was polished with alumina powder from Buehler (Dusseldorf, Germany), starting from 1.0 μm α Al_2O_3, down to 0.3 μm, and then 0.05 μm γ Al_2O_3. The graphite paper was cleaned with ethanol and later the surface was

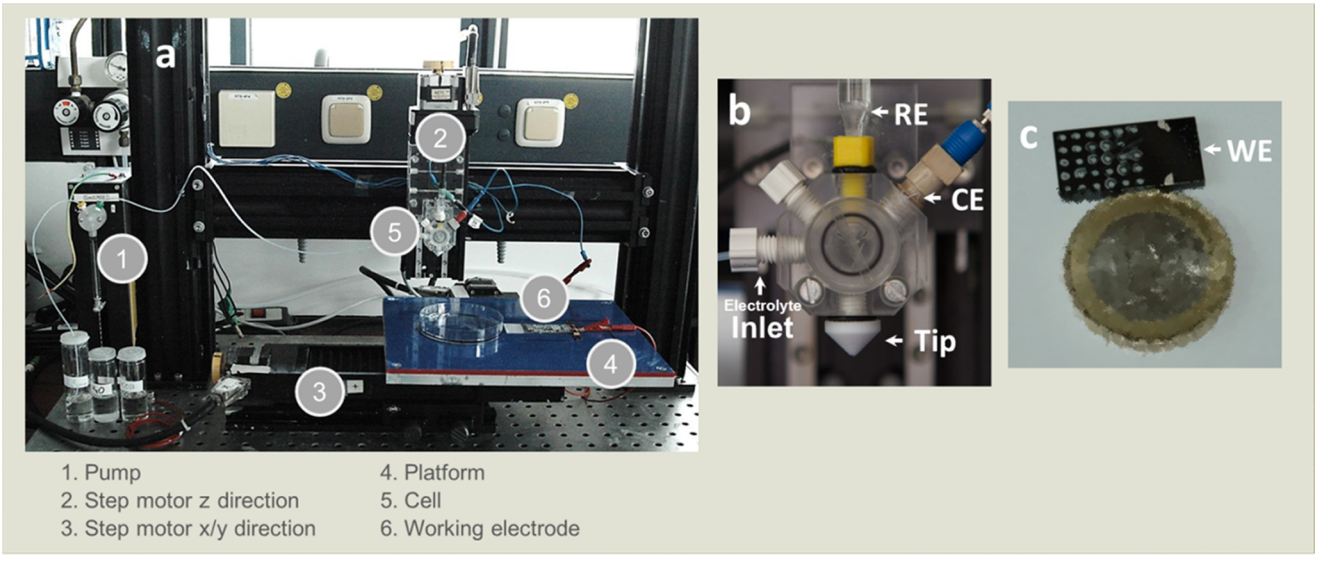


Figure 2. Photograph of a SDC system. (a) Entire setup including motors, connections, and pump. ((1) pump; (2) step motor for the z direction; (3) step motors for x/y directions; (4) sample platform; (5) electrochemical cell; (6) working electrode). (b) Electrochemical cell (or head), which contains the reference and counter electrode as well as an inlet for pumping the electrolyte. A Teflon tip is located at the bottom of the head. The opening of the tip (1 mm in diameter) determines the size of the area of the working electrode. (c) Glassy carbon plate used as working electrode showing 30 small Zn deposits.

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homogenized by polishing with 0.05 μm γ Al_2O_3. In both cases, copper tape was stuck to the WE to ensure proper electrical connection.

2.2. SEM and AFM Characterization. The morphology changes of the zinc deposits were investigated using a scanning electron microscope (Dual Beam TM 3D FEG; FEI). During the measurements, the chamber was maintained at about 6×10^{-10} mbar and operated at 20 kV. AFM (NanoWizard 3 from JPK Instruments) operating in tapping mode in air was used to investigate the topography.

2.3. Evaluation of Commercial Zn−Ni Batteries. AAA rechargeable Ni−Zn batteries (Powergenix) were tested as received. Galvanostatic chronopotentiometric measurements of commercial Ni−Zn batteries were carried out using a Bio-Logic VMP-3 potentiostat (Bio-Logic). PowerGenix AAA batteries can store 900 mWh, which results in a discharge capacity of 545 mAh for a cell voltage of 1.65 V. As the best performance of Ni−Zn batteries is achieved at a C-rate between 1 and 0.5 C, a charging current of 400 mA was chosen. The voltage range was −1.2/−2.0 V. CE and RE were short-circuited. The negative side acted as WE and the positive one as CE.

3. RESULTS AND DISCUSSION

3.1. High-Throughput Study of Zn Electroplating. An SDC was chosen as the high-throughput electrochemical technique (Figure 2a) to evaluate a large number of variables within a reasonable time frame. In short, an electrochemical cell is formed by pressing a Teflon orifice on a current collector to highly reproducibly select a defined electrode area. The selected area is then the working electrode at which any sequence of electrochemical processes can be performed. This type of variable in situ generated microcell design has been previously used in other fields such as corrosion of materials or photoelectrocatalysts.[24−26] The SDC not only accelerates the acquisition of experimental data but also minimizes any human error due to a high degree of automatization of all experiments. Another advantage of this setup is the small area used for each measurement (Figure 2c). Many samples can be prepared on a small area of a current collector hence facilitating fast alternative characterization by microscopy techniques.

Chronoamperometry using a glassy carbon plate as the working electrode (Figure 3) showed that the reduction of

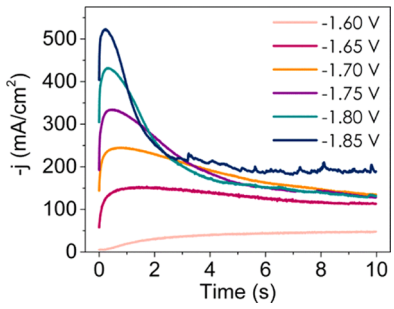


Figure 3. Chronoamperometric current transients for the electrodeposition of Zn onto a glassy carbon plate using an SDC at different applied potentials in 12 M KOH containing 1 M $Zn(OH)_4^{2-}$ ions.

$Zn(OH)_4^{2-}$ occurs below a potential of −1.60 V (vs Ag/AgCl/3 M KCl) in 12 M KOH containing 1 M $Zn(OH)_4^{2-}$. At low overpotentials, the reaction proceeds under a kinetically controlled regime. Hence, the concentration of $Zn(OH)_4^{2-}$ at the electrode surface is never completely depleted. The current density increases slightly over time due to the slowly increasing surface area. At high overpotentials, rapid electroplating of Zn leads to a fast increase in surface roughness, which concomitantly leads to a fast increase in current density. The fast consumption of $Zn(OH)_4^{2-}$ ions depletes their concentration close to the surface leading to control of the deposition by diffusion of the zincate ions to the electrode surface. Hence, the time necessary to reach a diffusion-controlled regime depends on the applied overpotential (Figure 3).

Ideally, the growth of Zn dendrites could be prevented by continuous electroplating establishing a kinetically controlled regime. In this scenario, neighboring sites do not need to compete for zincate ions, resulting in a homogeneous film. However, in practice this is not possible for two reasons. First, electroplating under kinetic control requires the application of small overpotentials or small current densities leading to slow plating rates. Hence, long charging times for a battery would result. Second, reduction of zincate ions occurs outside the stability window of water. Despite the hydrogen evolution reaction (HER) at Zn electrodes being kinetically slow enabling the kinetically faster reduction of the zincate ions, slow charging rates at small current densities enhance the thermodynamically favored but kinetically hindered HER. As a result, most of the charge during charging would be consumed by the HER instead of the reduction of zincate ions. Therefore, high current densities are desired to avoid parasitic HER and favor the reduction of zincate ions.

On the other hand, high current densities result in diffusion-controlled electroplating, which in turn promotes formation of Zn dendrites. As it takes a certain time to reach a diffusion-controlled regime (Figure 3), high current densities may be applied for a short time before the complete depletion of zincate ions at the surface takes place. After such a short current pulse at a high current density, the reaction can be stopped to allow the re-establishment of the concentration of zincate ions, thus avoiding a diffusion-controlled regime. Consequently, both the HER and formation of dendrites could be suppressed by applying short current pulses (Figure 4).

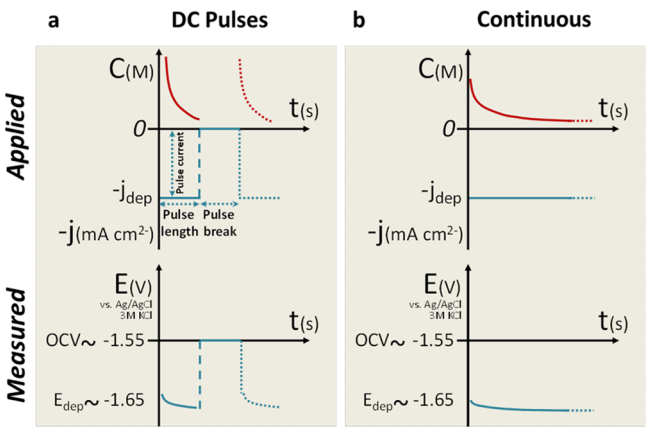


Figure 4. Schematic illustration of Zn electrodeposition by a (a) pulsed DC and (b) continuous protocol. Potentials are given versus Ag/AgCl/3 M KCl.

3.2. In Situ Roughness Evaluation of Zn Deposits. Several parameters, such as the amplitude and duration of the electroplating current pulse, and the duration of the relaxation pulse or of predeposition pulses, have to be systematically evaluated. This requires the investigation of a large number of samples (>1000 including replicates), making it practically impossible to analyze the roughness of each of these samples by SEM or AFM within a reasonable time frame. We hence chose

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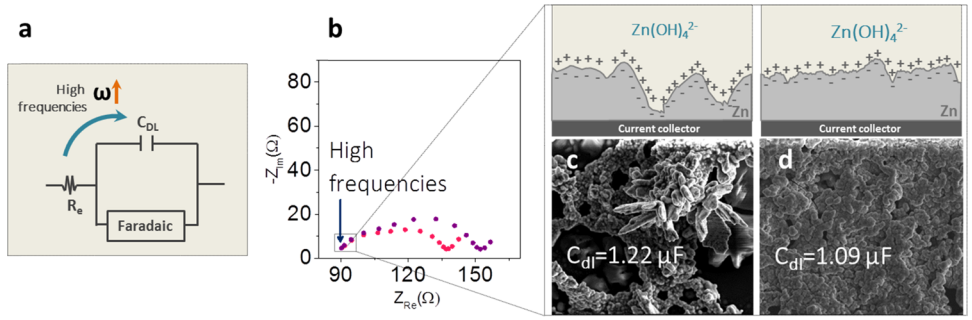


Figure 5. (a) Electric equivalent circuit and (b) Nyquist plot of EIS measurements of the Zn films electrodeposited on glassy carbon electrodes. SEM images of Zn films deposited by (c) one continuous pulse of −50 mA cm^{-2} and (d) a pulsed DC protocol (5/5 s, on/off, −50 mA cm^{-2}). The same charge of 7.5 C cm^{-2} was employed for both films. The sketches shown over the SEM images relate the capacitance values measured by EIS with the roughness of the electrode surface. The smoother sample surface, the lower the capacitance value.

EIS as a high-throughput electroanalytical method for the in situ evaluation of the surface roughness after electroplating.

EIS provides information about electrochemical processes occurring at different time constants by applying a sinusoidal perturbation voltage superimposed on a constant applied potential. When the applied frequencies are higher than the time constant of any of the Faraday processes, the overall impedance is dominated by the resistance of the electrolyte and the capacitance of the electrochemical double layer at the electrode interface (Figure 5a). As the capacitance is directly proportional to the specific capacitance, which is a constant value, and the surface area, it can be used to evaluate changes in the surface area. The capacitance of the electrochemical double layer, CDL, at the electrode interface can be determined by[27]

$$C_{DL} = -\frac{1}{2\pi f Z_{Im}} \quad (1)$$

Two simple electroplating conditions were initially chosen to explore the hypothesis. Continuous galvanostatic electroplating at a current density of −50 mA cm^{-2} for 150 s was used as the reference sample. The electroplating at −50 mA cm^{-2} was split into 30 pulses with a duration of 5 s inserting a resting and relaxation period of 5 s between two plating pulses. By this method, a comparable amount of Zn was deposited in both cases.

SEM images of the two samples clearly show the positive influence of the pulse electroplating. No dendrite formation was observed when applying the pulse protocol (Figure 5c,d). EIS was performed at OCP immediately after Zn deposition. Figure 5b shows the Nyquist plots of the Zn films. We chose a high frequency (21 kHz) to avoid on the one hand, start-up transients, and on the other hand, small leakage through the Faradaic path (Figure 5a). At 21 kHz, capacitances of 1.22 and 1.09 μF for continuous and pulsed electroplating were derived, respectively. The higher capacitance value is due to the higher surface area (Figure 5c,d). The same trend was observed at 46 and 10 KHz. In the following discussion, the value of the imaginary part of the impedance at 21−24 kHz is used as in situ derived descriptor of the surface roughness. It should be noted that the imaginary part of the impedance can only be used as an approximate descriptor as both a flat surface with only a few dendrites and a rough surface without dendrites lead to small values of capacitance. However, evidently both cases are detrimental to the performance of Zn electrodes. Hence, the primary goal of this work is to find Zn deposition protocols leading to smooth surfaces that exhibit a large imaginary value of impedance. SEM and AFM were additionally employed for selected samples to evaluate their morphology, and in turn, to confirm the in situ obtained EIS results.

3.3. Importance of Early Stages on the Deposition of Dendrite-Free Zn Electrodes. Graphite paper was used as the current collector due to its good adhesion to Zn deposits. As expected, electroplating on graphite paper by means of chronoamperometry at different potentials showed a similar behavior to that on glassy carbon electrodes (see Figure S1 compared with Figure 3). The diffusion-limited current after 10 s was around −55 mA cm^{-2} for the largest applied overpotentials, a value that is very close to the theoretical one of −60 mA cm^{-2}, as predicted by the Cottrell equation. In contrast, on the glassy carbon surfaces, current densities of about −150 mA cm^{-2} were observed, which are attributed to the additional contribution of the HER. Obviously, the HER is more favorable on glassy carbon surfaces than that on graphite paper.

A current density of −56 mA cm^{-2} was chosen for all galvanostatic electroplating experiments as it is a reasonable value for practical battery applications. For example, a Zn electrode with a mass loading of 50 mg cm^{-2} would require a current density of −30 mA cm^{-2} to be charged within 1 h (1 C). As shown above, a simple pulsed electroplating with 5/5 s (on/off) completely prevented Zn dendrite formation (Figure 5c,d). To further optimize the protocols for even smoother Zn surfaces, pulsed electroplating with a shorter pulse duration was investigated. No additional surface smoothing was found by decreasing the electroplating pulse from 5 to 2 s (Figure 6).

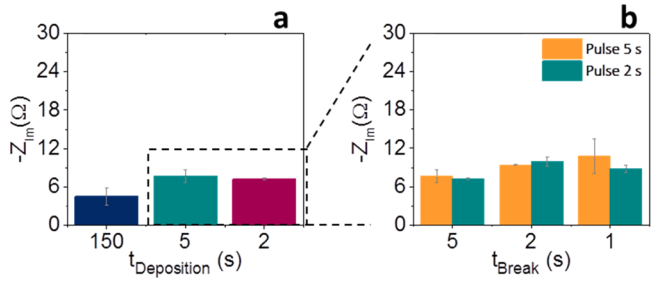


Figure 6. Imaginary part of impedance, $-Z_{Im}$, at a frequency of ∼24 kHz for Zn films deposited at different conditions. (a) Influence of the duration of the electroplating pulse (2 s relaxation break). (b) Influence of the duration of the relaxation break. A current density of −56 mA cm^{-2} and a total charge of 8.4 C cm^{-2} were applied to all samples (n > 10).

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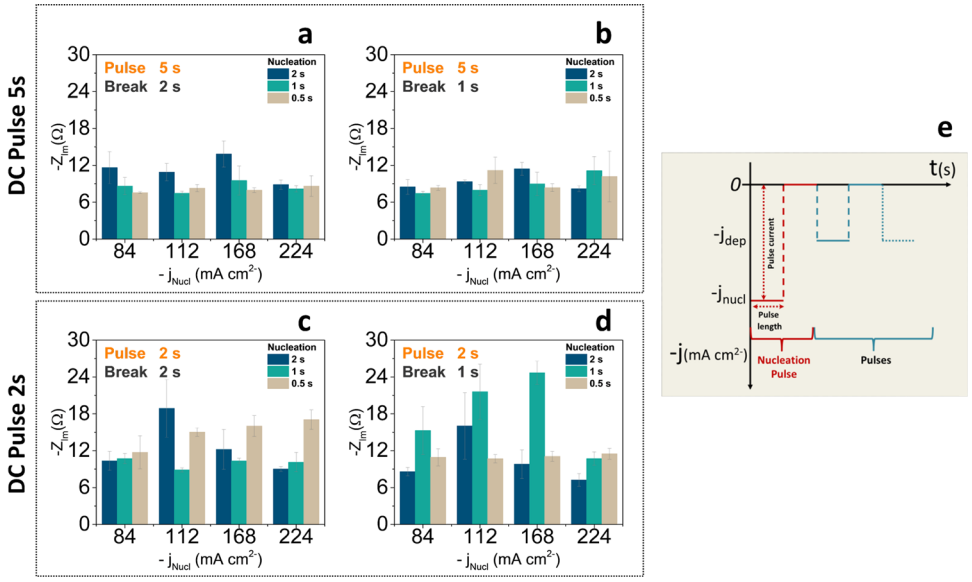


Figure 7. Imaginary part of the impedance, $-Z_{Im}$, at a frequency of ~24 kHz at Zn films obtained under different electroplating conditions. The nucleation step is applied only once before the pulsed charging protocol: a nucleation step of −84, −112, −168, and −224 mA cm^{-2} for 0.5, 1, or 2 s followed by a pulsed DC charging protocol of −56 mA cm^{-2}: 5/2, 5/1, 2/2, and 2/1 s (on/off) for (a), (b), (c), and (d), respectively. The same charge (8.4 C cm^{-2}) was consumed for the electrodeposition of all films. A table giving detailed information about the measurements is included in the Supporting Information. (e) Schematic representation of a nucleation step followed by the pulsed DC charging protocol (n > 5).

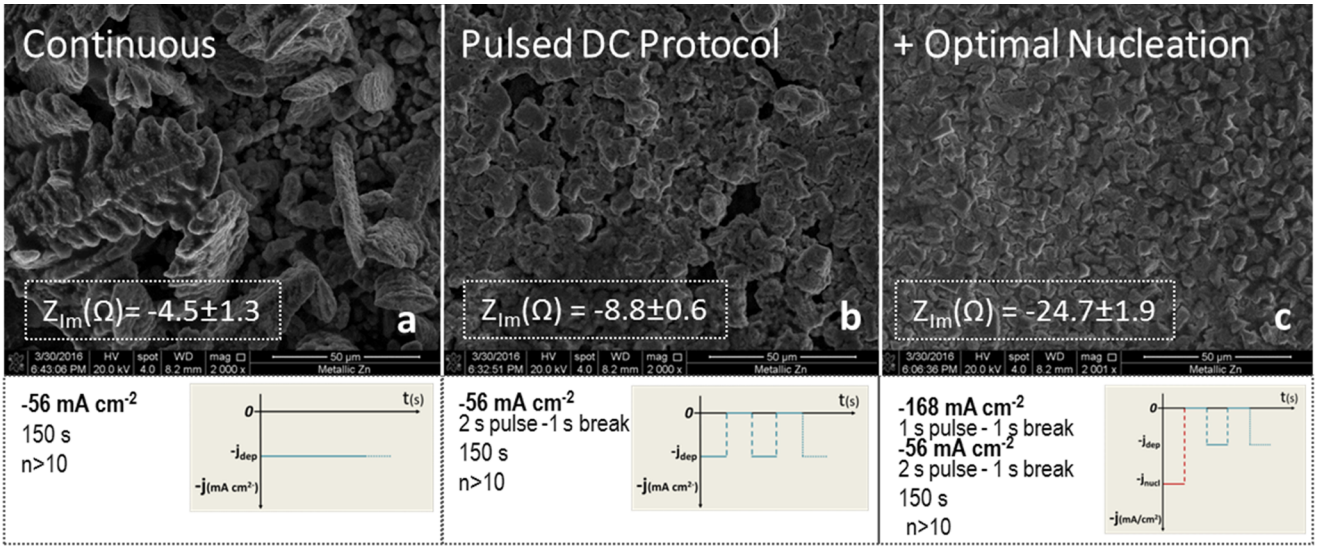


Figure 8. SEM images of Zn films electrodeposited on graphite paper from a solution of 12 M KOH containing 1 M $Zn(OH)_4^{2-}$ by applying (a) one continuous charging pulse of 150 s at −56 mA cm^{-2}, (b) a pulsed DC (2/1 s, on/off, −56 mA cm^{-2}) protocol, and (c) a nucleation step at −168 mA cm^{-2} for 1 s followed by a pulsed DC (2/1 s, on/off, −56 mA cm^{-2}) protocol. The electrodeposition conditions are given and are schematically illustrated below each SEM image. The value of $-Z_{Im}$ from EIS is also provided (n > 10).

Also, the duration of the resting period between two subsequent electroplating pulses did not have a drastic impact on the roughness of the Zn deposits. Interestingly, relaxation periods of 1 s resulted in smoother films as compared to those that had 5 and 2 s, indicating that a resting time of 1 s between subsequent electroplating pulses is sufficiently long to re-establish the $Zn(OH)_4^{2-}$ ion concentration at the electrode surface.

To further decrease the roughness of the deposited Zn films, an additional step was included in the charging protocol. From studies on the nucleation and growth of metals, it is known that the early stages of the electrochemical deposition are of high importance for the further growth of the metal deposit.[28] The number of active sites at which nuclei grow increases with increasing overpotential. To achieve a high coverage of Zn, which prevents the HER and the creation of a few preferential metal deposition sites leading in turn to the growth of dendrites, a short current pulse with a high current density and concomitantly a high overpotential was implemented at the beginning of the charging protocol (Figure 7e). After the initial short and high amplitude electroplating pulse, which is essentially a nucleation pulse, the charging protocol was continued with current pulses of lower amplitude (−56 mA cm^{-2}).

The influence of the duration and amplitude of the nucleation pulse, which was applied only once at the beginning

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of the charging protocol, on the roughness of the deposited Zn film was evaluated using in situ EIS. The results are summarized in Figure 7a–d and correspond to the influence of the nucleation pulse on the subsequent 5/2, 5/1, 2/2, and 2/1 s (on/off) pulsed DC protocols, respectively. The highest value of the imaginary part in the corresponding in situ EIS ($-Z_{Im}$ = 24.7 ± 1.9 Ω) was achieved for a nucleation pulse with a duration and amplitude of 1 s and −168 mA cm^{-2}, respectively, followed by a 2/1 s (on/off) pulsed DC protocol with an amplitude of −56 mA cm^{-2}. The implementation of a short nucleation pulse applied only once at the beginning of the pulsed DC charging protocol (2/1, on/off) led to a 3-fold decrease in the average roughness and a 5.5-fold decrease as compared to that from continuous electroplating without any relaxation periods

3.4. SEM and AFM. As the roughness values obtained from EIS data are only approximate descriptors, SEM and AFM of selected samples were used to validate the results of the high-throughput in situ EIS measurements. Three samples were selected, namely, one obtained by continuous electroplating, one from a 2/1 s (on/off) pulsed DC protocol (−56 mA cm^{-2}), and one with an additional nucleation pulse (1 s at −168 mA cm^{-2}) followed by the 2/1 s (on/off) pulsed DC protocol (−56 mA cm^{-2}). The SEM results were consistent with the EIS data (Figure 8). Dendrites were clearly visible for the sample obtained by continuous electroplating. The formation of dendrites was largely prevented by applying a 2/1 s (on/off) pulsed DC protocol. However, the electrode surface was not completely uniform as discontinuities in the Zn film, such as empty gaps, could be observed. Introducing a short nucleation pulse at higher current density before the 2/1 s (on/off) pulsed DC protocol led to an apparently smoother film without any visible discontinuities.

As SEM shows only a two-dimensional representation of the morphology, AFM was additionally employed to provide the required three-dimensional information and thus to confirm the benefits of the nucleation pulse in achieving smooth Zn deposits. AFM measurements were also attempted on the continuously electroplated Zn films, however, due to the presence of dendrites, the roughness of these samples exceeded the limits of our microscope (max. z-height of 8 μm). Figure 9 shows the AFM images of the films deposited using pulsed DC protocols with and without the nucleation pulse. AFM measurements confirmed that introducing a short nucleation step resulted in smoother films in addition to preventing discontinuities in the film (Figure 8).

3.5. Influence of the Pulsed DC Charging Protocol on Commercial Rechargeable Ni–Zn Batteries. The previous results demonstrate that the formation of dendrites and roughening of the electrode surface during charging can be prevented by applying pulsed DC protocols comprising relaxation periods between short electroplating pulses. As the prevention of dendrite formation at Zn electrodes should be beneficial for the long-term performance and especially for the cyclability of Ni–Zn batteries, commercially available AAA Ni–Zn batteries from PowerGenix (Figure 10a) were purchased and used to explore the impact of a pulsed DC charging protocol on their cyclability. The commercial AAA Ni–Zn batteries can store 900 mWh, which results in a discharge capacity of 545 mAh at a cell voltage of 1.65 V.

As the AAA Ni–Zn batteries are recommended by the manufacturer to be charged at a C-rate between 1 and 0.5 C, a charging current of 400 mA was chosen. It should be noted that

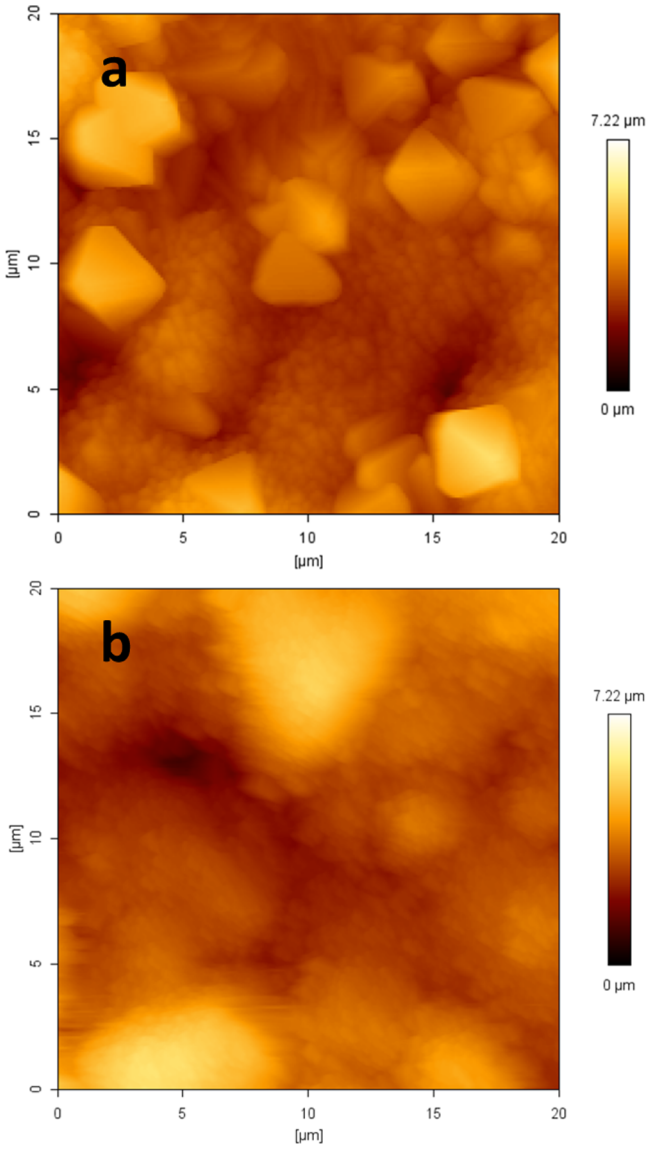


Figure 9. AFM images of Zn films deposited on graphite paper from a solution of 12 M KOH containing 1 M $Zn(OH)_4^{2-}$ ions by applying (a) pulsed DC protocol (2/1 s, on/off, at −56 mA cm^{-2}) and (b) an initial nucleation step (1 s at −168 mA cm^{-2}) followed by the pulsed DC protocol (2/1 s, on/off, at −56 mA cm^{-2}).

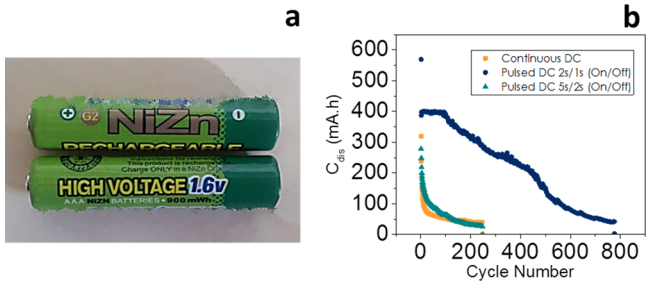


Figure 10. (a) Photograph of two commercial Ni–Zn AAA batteries from PowerGenix. (b) Discharge capacity of PowerGenix AAA Ni–Zn batteries charged using continuous (orange) and pulsed DC protocols: (2/1 s, on/off) (blue), (5/2 s, on/off) (turquoise) protocols at a current of 400 mA (nucleation step was not applied).

all given voltage values are positive because the Ni electrode is connected as the working electrode. The voltage range was

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−2.0/−1.2 V, except for the first cycle in which the upper cutoff was increased to −2.2 V (see voltage profiles in the Supporting Information). This was necessary due to the phase transformation at the Ni electrode during the first cycle.[29] Oxygen evolution at the Ni electrode during the first cycle was responsible for the observed lower Coulombic efficiency in this cycle (see Figure S3).[30] Figure 10b shows the charge capacity obtained from a commercial Ni−Zn battery when applying the continuous and pulsed DC (5/2 s, on/off) and pulsed DC (2/1 s, on/off) charging protocols (without the nucleation step).

After continuous charging, the discharge capacity rapidly dropped below 25% of the theoretical value (136 mAh) during the first 10 cycles. It should be noted that several new batteries were tested and all showed similar performance. After applying the pulsed DC (2/1 s, on/off) charging protocol, the discharge capacity remained above 50% (272 mAh) and 25% (136 mAh) of the theoretical value for 275 cycles and >500 cycles, respectively. These results clearly demonstrate that the prevention of dendrite formation and the electroplating of smoother Zn films achieved by pulsed DC charging protocols lead to improved performance, even for commercial batteries based on Zn anodes. However, no improvement was observed when applying the longer pulsed DC (5/2 s, on/off) charging protocol due to the fact that the duration of the pulse (5 s) was too long. The lack of improvement for this pulsed DC protocol clearly shows the importance of optimizing the charging protocol for the commercial device.

4. CONCLUSIONS

The formation of undesired Zn dendrites during electroplating under a diffusion-controlled regime occurs due to the enhanced mass transport at sharp features with respect to flat surfaces. The introduction of short resting periods during the electroplating of Zn prevents the formation of Zn dendrites, resulting in smoother films as the concentration of zincate is re-established at the electrode surface avoiding the competition of neighboring sites for zincate.

An SDC containing a solution of 12 M KOH + 1 M $Zn(OH)_4^{2-}$ allowed the exploration of many different pulsed DC protocols to find the optimal Zn electroplating conditions. The introduction of a nucleation pulse (short pulse of large amplitude), which allows the achievement of a high initial coverage of Zn nuclei, appears to be more important than the length of the plating and resting pulses. The combination of an optimal nucleation step and pulsed DC charging protocol results in smooth Zn films, completely preventing the formation of dendrites and discontinuities in the film.

Commercially available Ni−Zn batteries such as PowerGenix AAA (900 mWh) benefit from pulsed DC charging protocols. The cyclability of a PowerGenix battery can be prolonged from a few dozen cycles to several hundreds of cycles when (dis)charged at 400 mAh. Therefore, the insights gathered for Zn electroplating using a model system are valid for commercial Zn anode-based batteries.

■ ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge on the ACS Publications website at DOI: 10.1021/acsami.7b01705.

Table showing the record of the imaginary part of impedance, $-Z_{Im}$, at a frequency of ∼24 kHz at Zn film obtained at different electroplating conditions; figure corresponding to the potentiostatic current density transients for the deposition of Zn in graphite paper; two figures showing the charge−discharge curves for the galvanostatic cycling with potential limitation of Ni−Zn PowerGenix AAA cells (PDF)

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Notes

The authors declare no competing financial interest.

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